Representations of Plurality in LLMs

UU NLP Group Meeting

February 5th, 2026

ANH DANG

Utrecht University

Questions:

- How do LLMs internally represent the antecedents of a plural entity?
- How do LLMs decide the antecedents for a plural reference?

Yesterday, John and Mary came to the shop. They bought some milk.

- **How do LLMs know that "They" refer to John and Mary?**
- **Where do they store such information?**
- **Do they track and update coreference information throughout discourse?**

The Geometry of Truth: Emergent Linear Structure in LLM Representations of True/False Datasets

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Abstract

Large Language Models (LLMs) have impressive capabilities, but are prone to outputting falsehoods. Recent work has developed techniques for inferring whether a LLM is telling the truth by training probes on the LLM's internal activations. However, this line of work is controversial, with some authors pointing out failures of these probes to generalize in basic ways, among other conceptual issues. In this work, we use high-quality datasets of simple true/false statements to study in detail the structure of LLM representations of truth, drawing on three lines of evidence: 1. Visualizations of LLM true/false statement representations, which reveal clear linear structure. 2. Transfer experiments in which probes trained on one dataset generalize to different datasets. 3. Causal evidence obtained by surgically intervening in a LLM's forward pass, causing it to treat false statements as true and vice versa. Overall, we present evidence that at sufficient scale, LLMs *linearly represent* the truth or falsehood of factual statements. We also show that simple difference-in-mean probes generalize as well as other probing techniques while identifying directions which are more causally implicated in model outputs.

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Principal Component Analysis

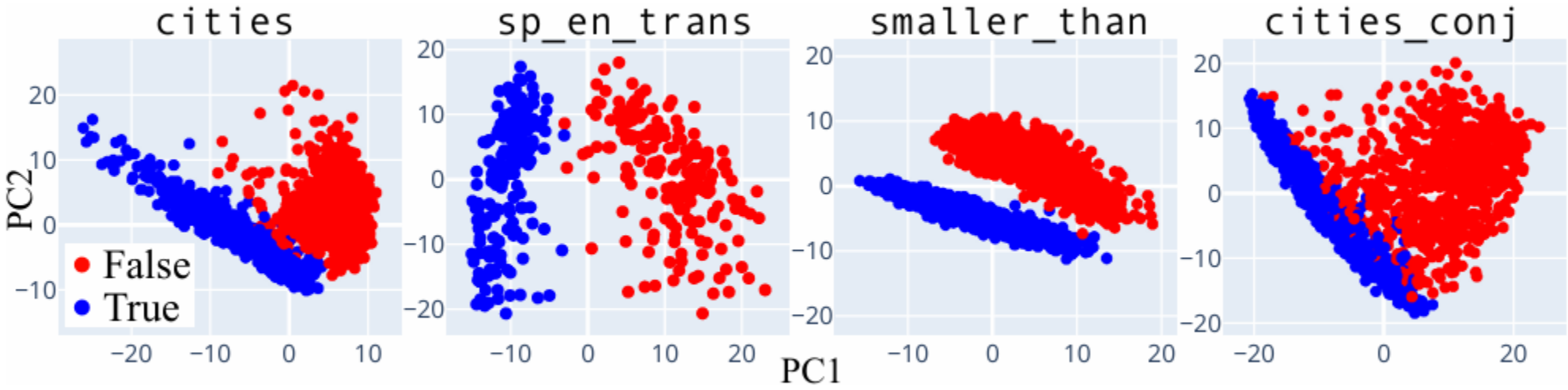

Figure 1: PCA visualizations for LLaMA-2-70B representations of our true/false datasets.

`cities`	"The city of `[city]` is in `[country]`."
`neg_cities`	Negations of statements in `cities` with "not"
`sp_en_trans`	"The Spanish word '`[word]`' means '`[English word]`'."
`neg_sp_en_trans`	Negations of statements in `sp_en_trans` with "not"
`larger_than`	"x is larger than y."
`smaller_than`	"x is smaller than y."

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Representational Analysis of Binding in Language Models

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Abstract

Entity tracking is essential for complex reasoning. To perform in-context entity tracking, language models (LMs) must bind an entity to its attribute (e.g., bind a container to its content) to recall attribute for a given entity. For example, given a context mentioning "The coffee is in Box Z, the stone is in Box M, the map is in Box H", to infer "Box Z contains the coffee" later, LMs must bind "Box Z" to "coffee". To explain the binding behaviour of LMs, Feng and Steinhardt (2023) introduce a Binding ID mechanism and state that LMs use a abstract concept called Binding ID (BI) to

tributes (Feng and Steinhardt, 2023). For example, given Sample 1 and 2, a model must bind the entities (e.g., "Box Z", "Box M", "Box H", "Alex", "John" and "Carl") to their corresponding attributes (e.g., "coffee", "stone", "map", "bean", "pie" and "fruit") so as to recall (or answer) such as what is in "Box Z" or what is sold by "Alex" without confusion. Binding has also been studied as a fundamental problem in Psychology (Treisman, 1996).

To uncover how Language Models (LMs) realize binding in term of internal representation, Feng and Steinhardt (2023) introduce a Binding ID mechanism and state that LMs apply a abstract concept

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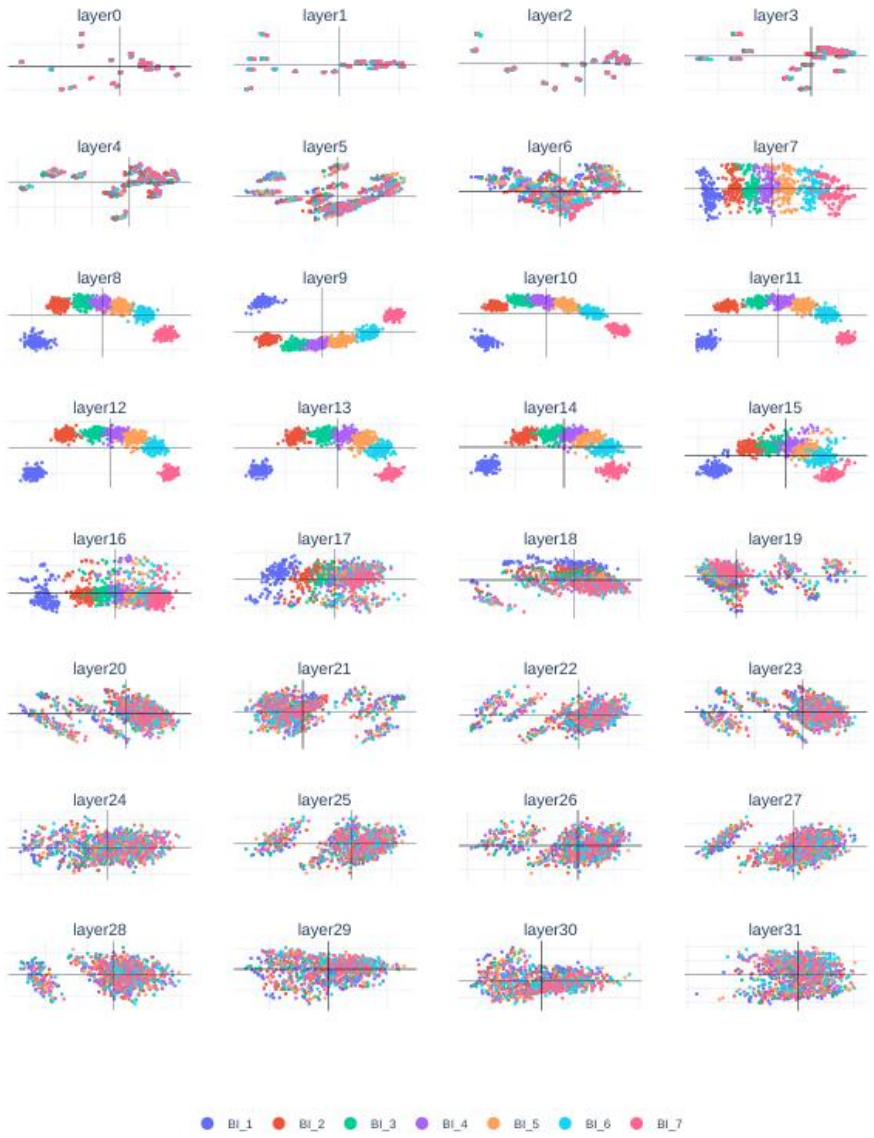

Figure 2: Layer-wise OI subspace visualization on Llama2-7B, where "BI" primarily denotes OI.

Principal Component Analysis

attribute entity

The **coffee** is in **Box Z**, the **stone** is in **Box M**, the **map** is in **Box H**. **Box Z** contains the __.

In an activation subspace, **entities with the same Ordering Index tend to cluster together,** even though these entities have **different semantic meaning**, and the OIs are encoded as directions in the subspace.

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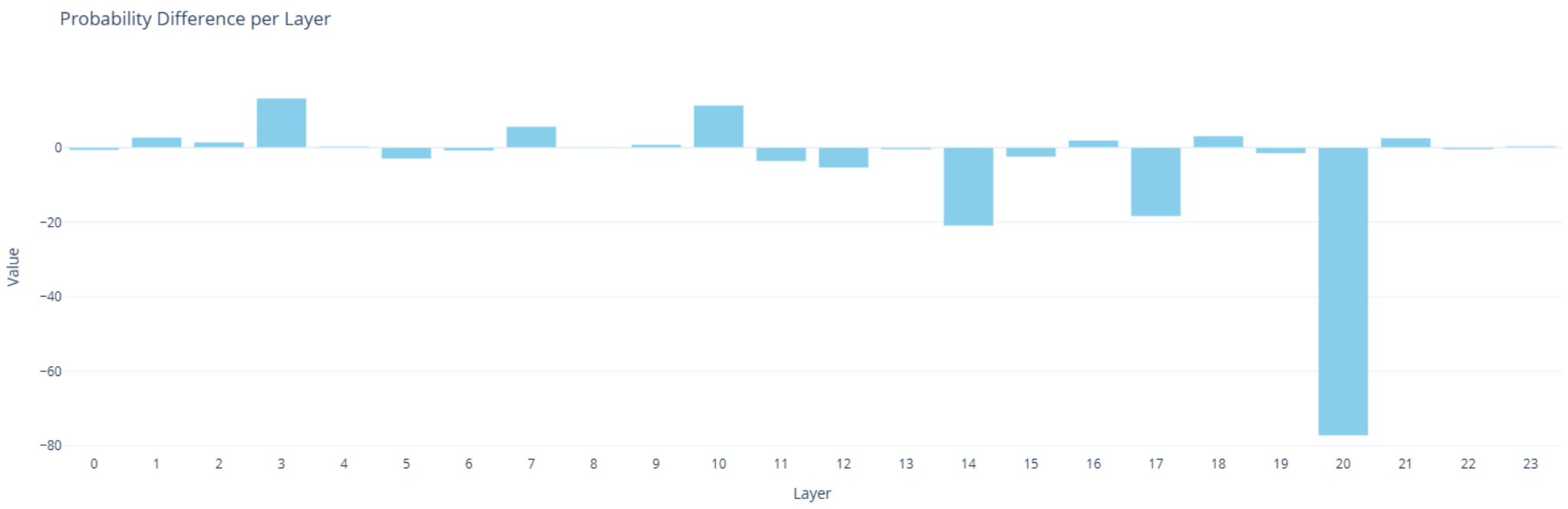
Activation Patching

Method: Activation Patching
Component: Residual Stream
Metric: Logit Difference

patching token

John saw Mary with her daughter and waved **at _them_**
John saw Mary with her bike and waved at **_her_**

Logit_Diff = L_{them} - L_{her}



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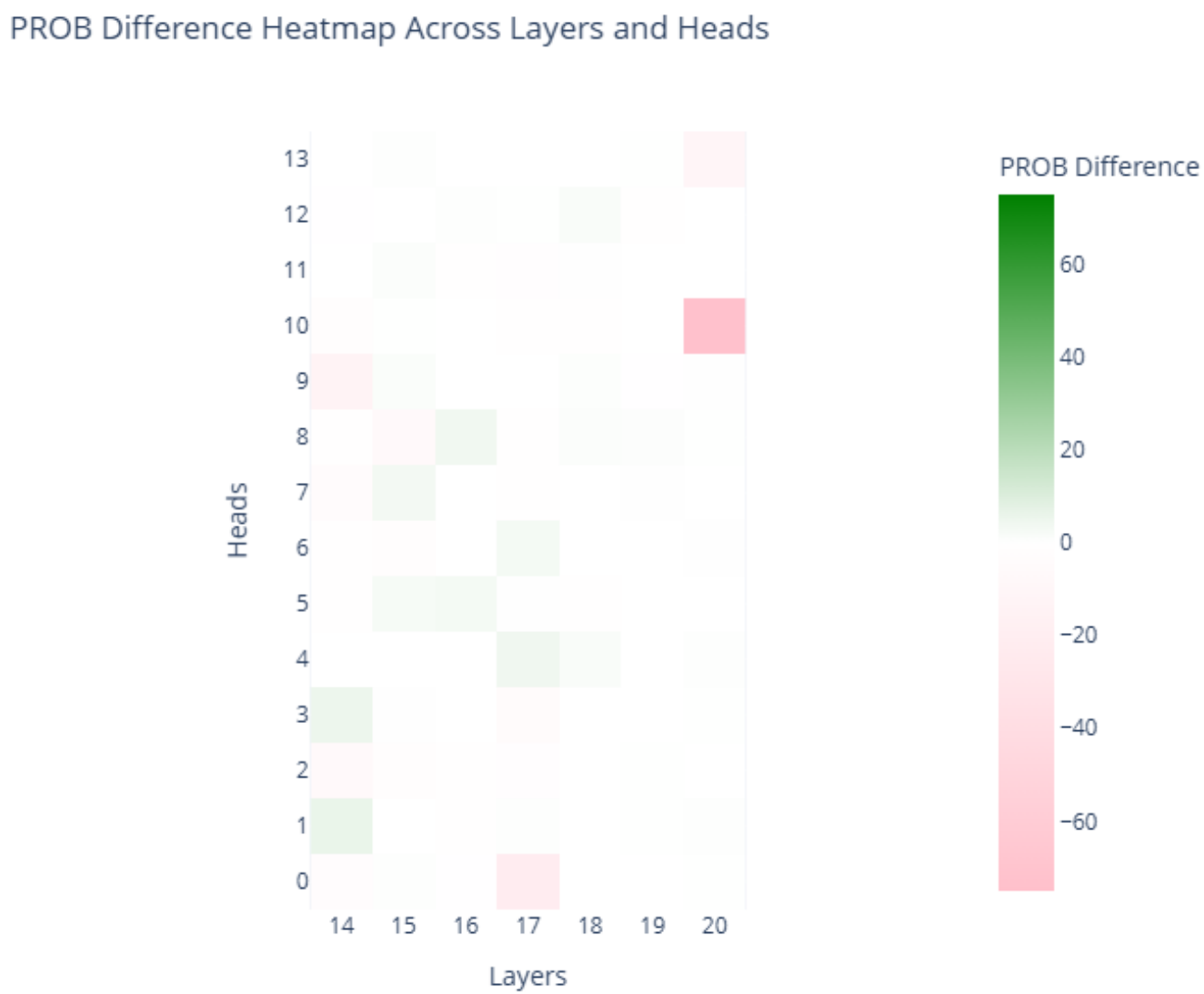

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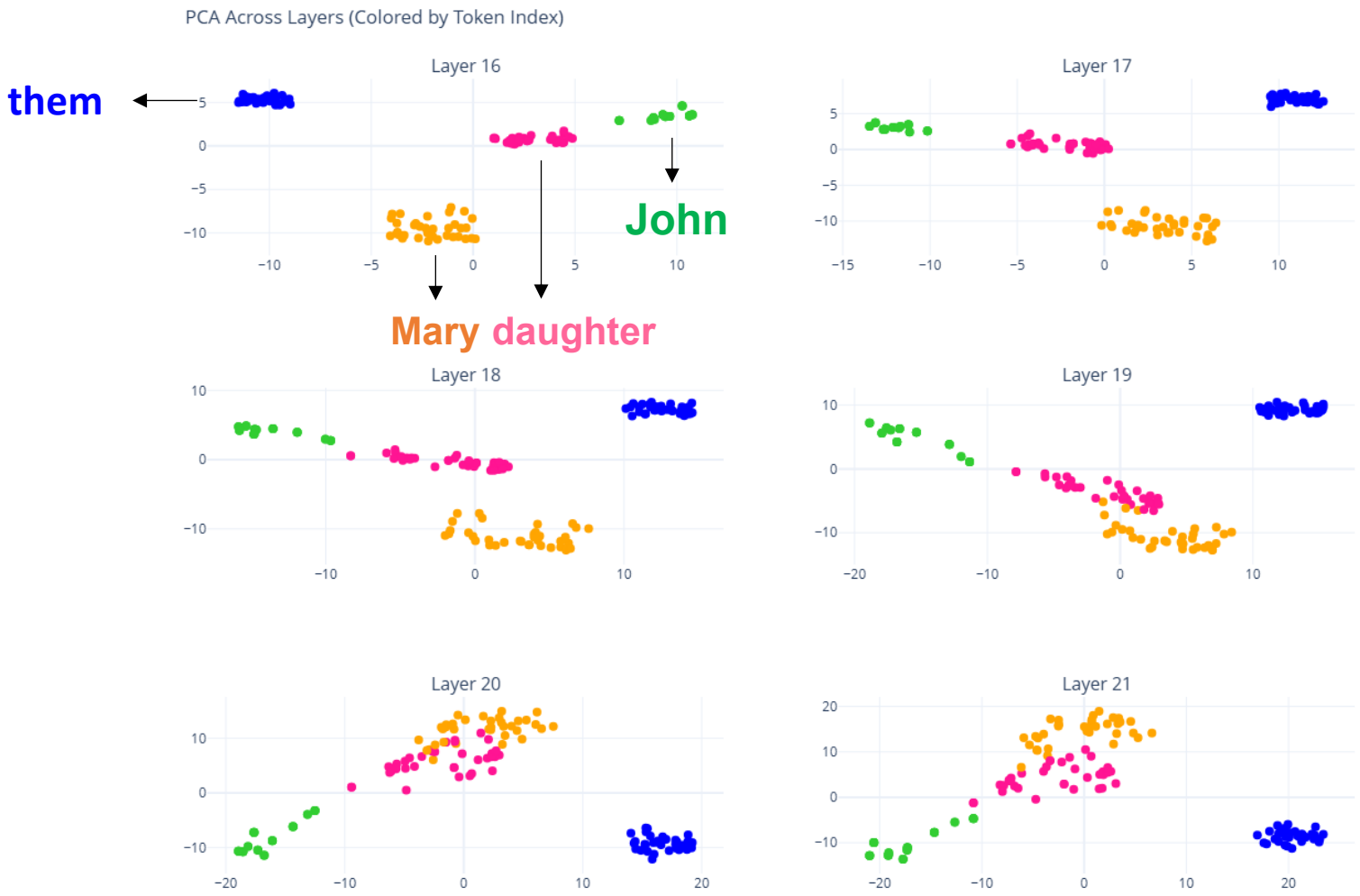
antecedents

John saw Mary with her daughter and waved at them



- LLMs dedicate **subspaces for token positions**
- **Overlapping between two antecedents** of the plural pronoun at later layers.

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John saw Mary with her bike and waved at them

- **No overlapping** observed when the plural pronoun **is not preferred**

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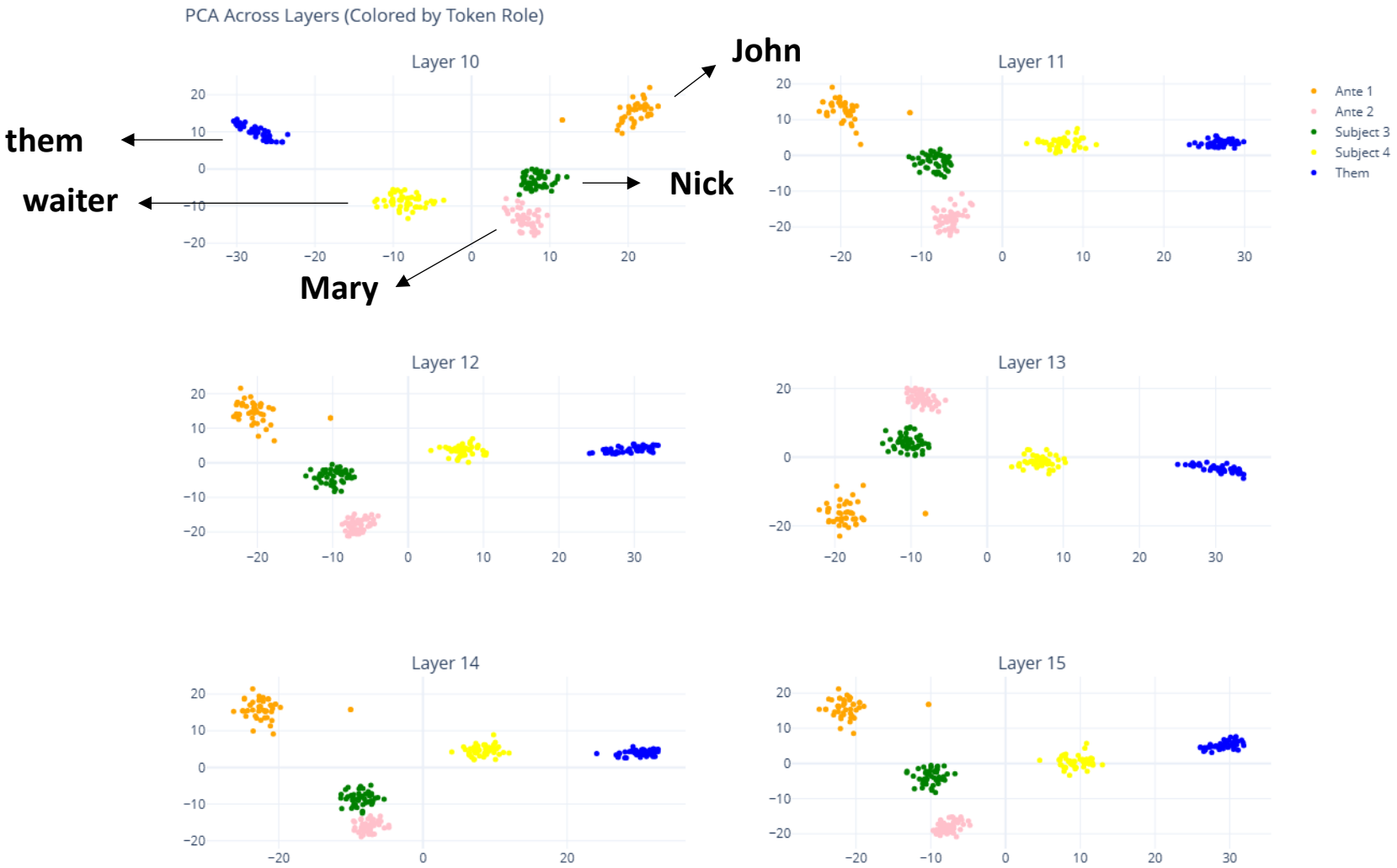
When John and Mary arrived at the restaurant that Nick owned, the waiter welcomed them



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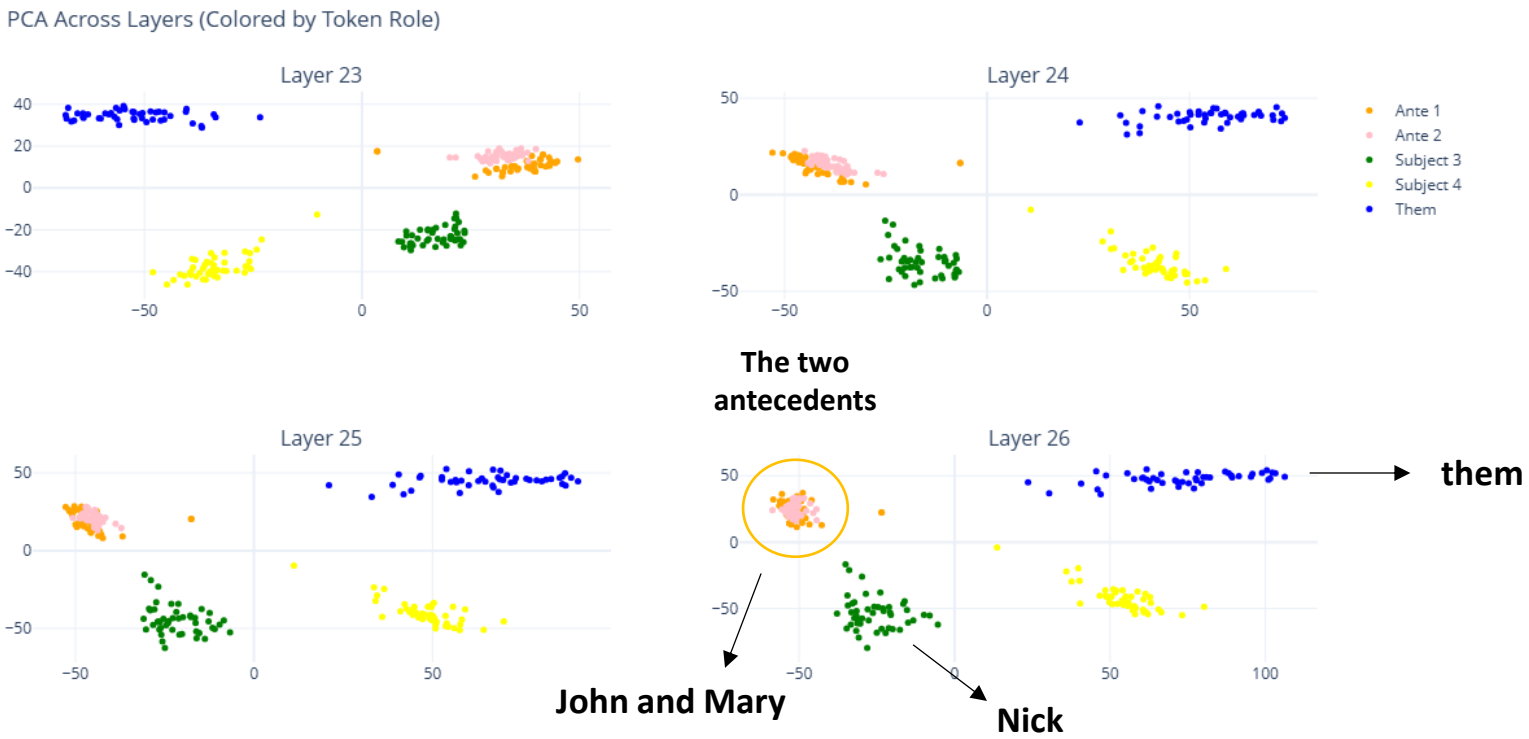
When **John** and **Mary** arrived at the restaurant that **Nick** owned, the **waiter** welcomed **them**



- The activations of the two antecedents (John and Mary) **overlaps along 3 PCs at later layers**
- **Subspaces for plural entities?**

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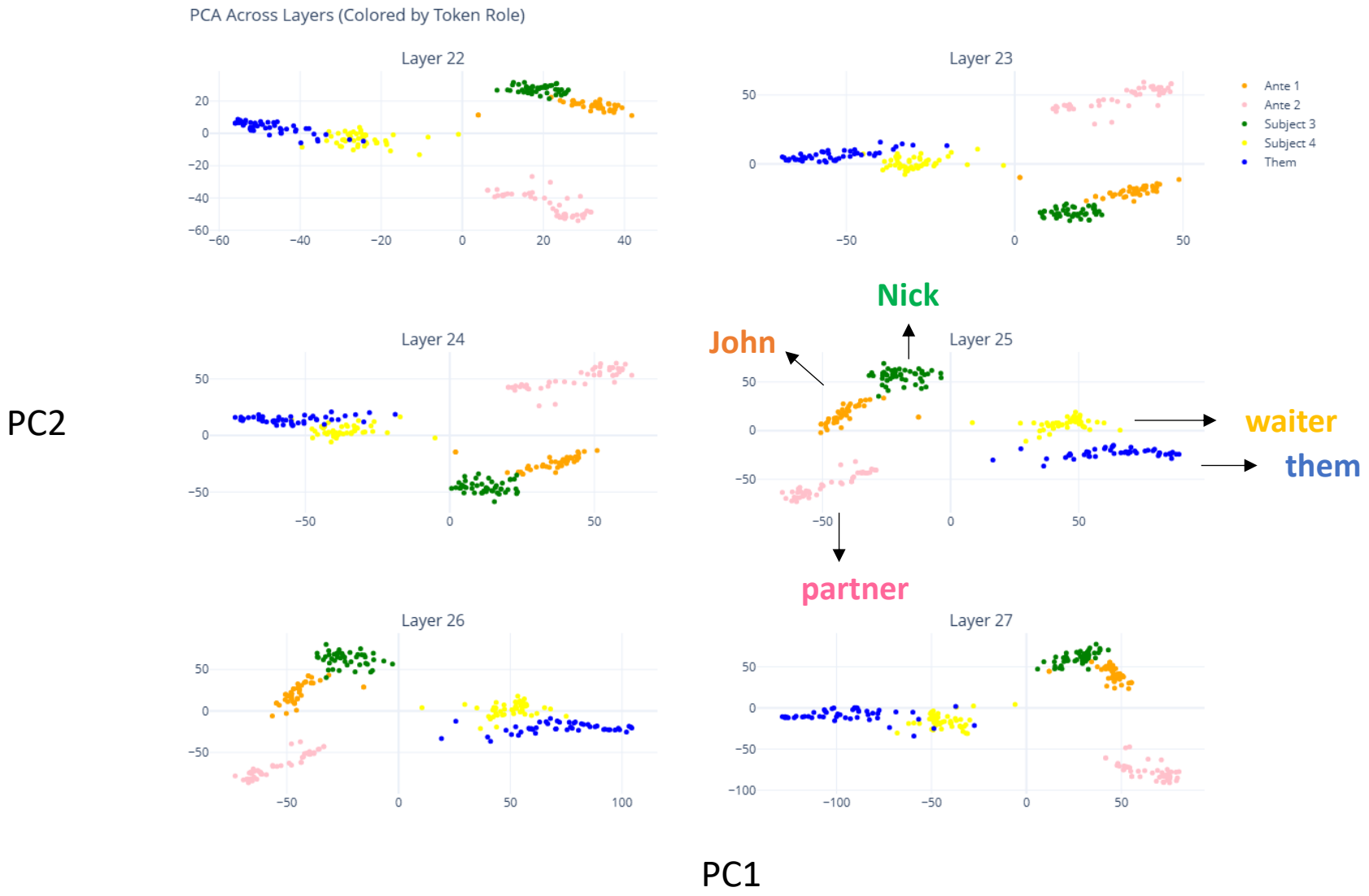
When **John** and his **partner** arrived at the restaurant that **Nick** owned, the **waiter** welcomed **them**



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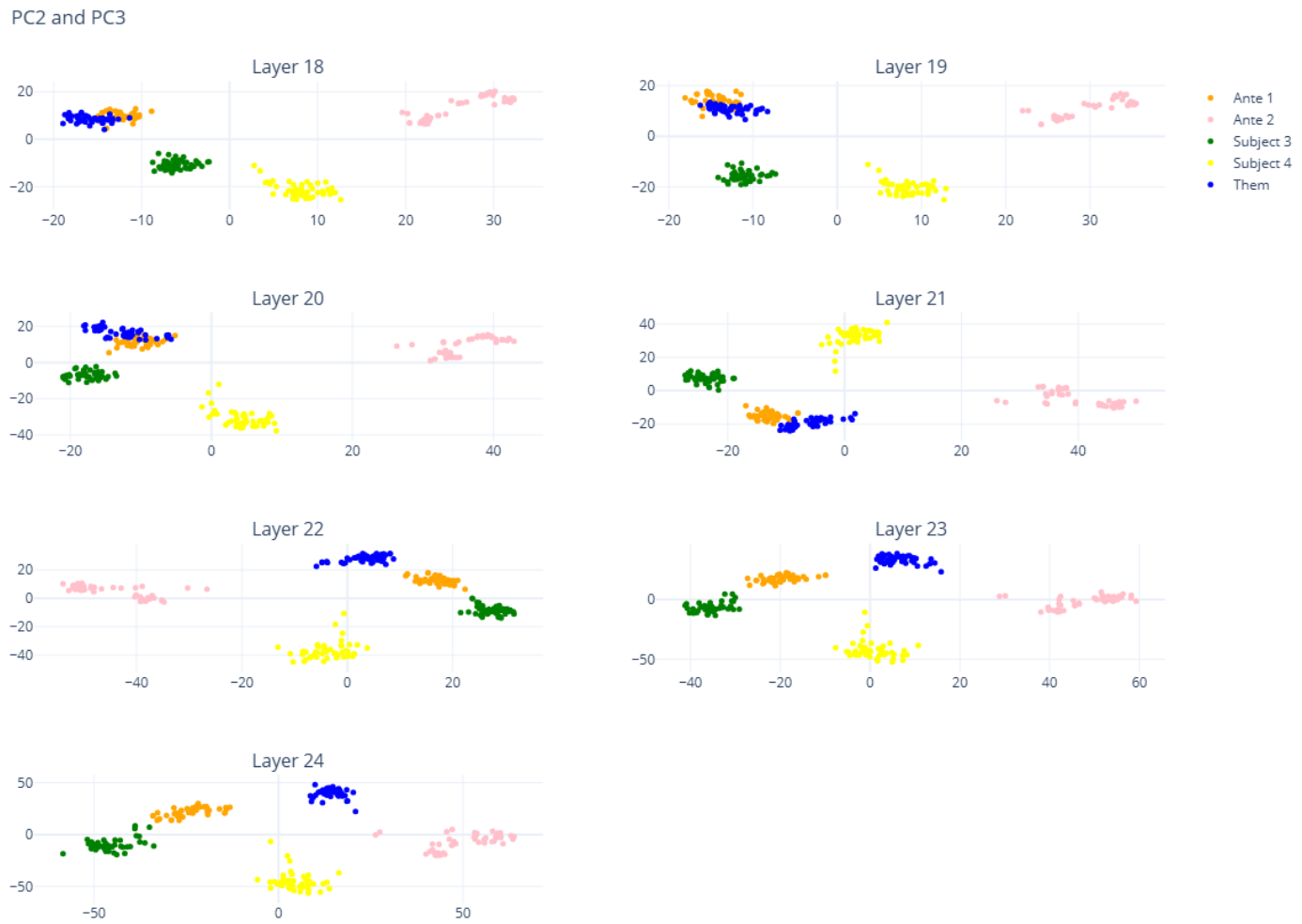
When **John** and his **partner** arrived at the restaurant that **Nick** owned, the **waiter** welcomed **them**



PC3

PC2

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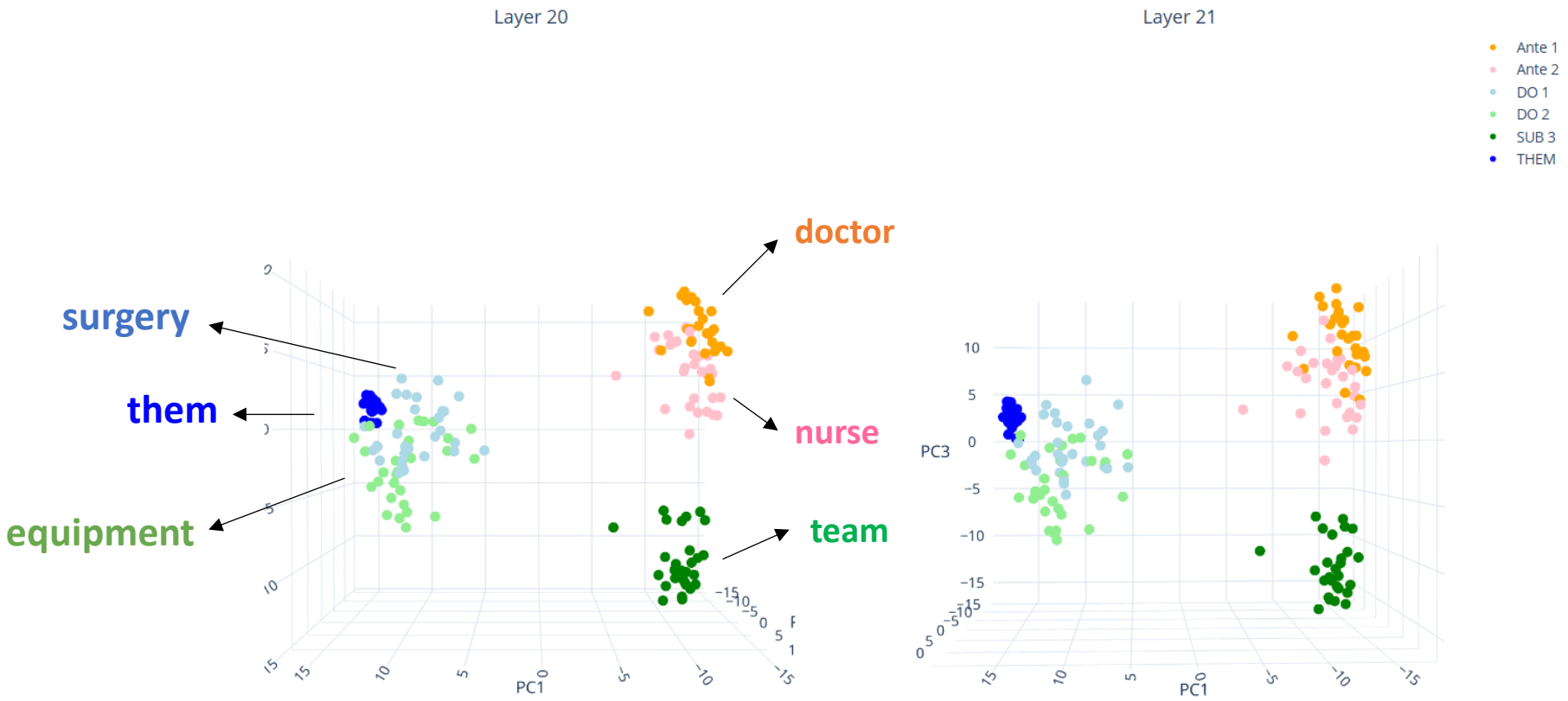
When the hospital lost electricity, the **doctor** was **performing surgery,** and the **nurse** was **preparing equipment**, the backup **team** came and rescued **them**



- **LLMs may dedicate subspaces for semantic roles**

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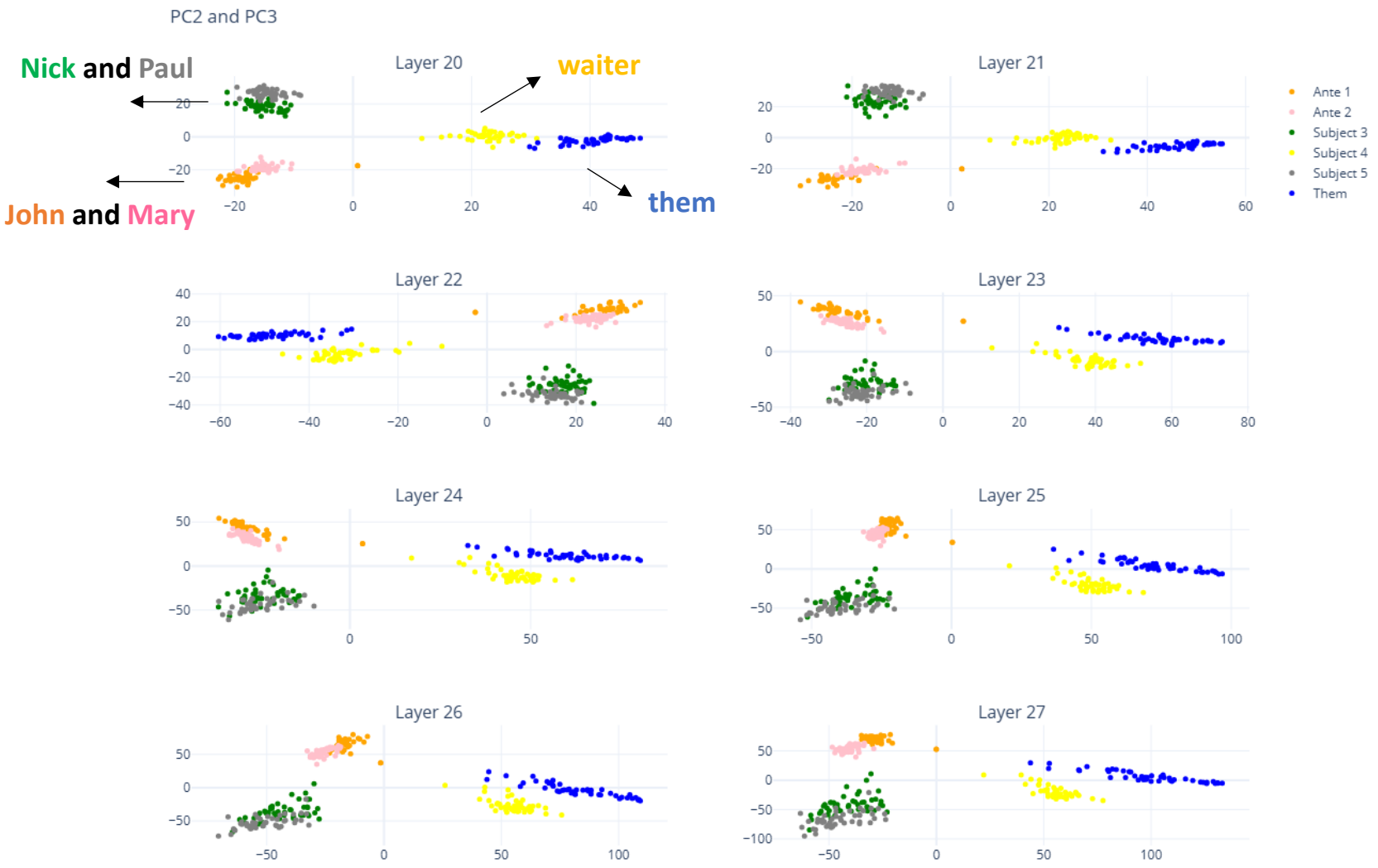
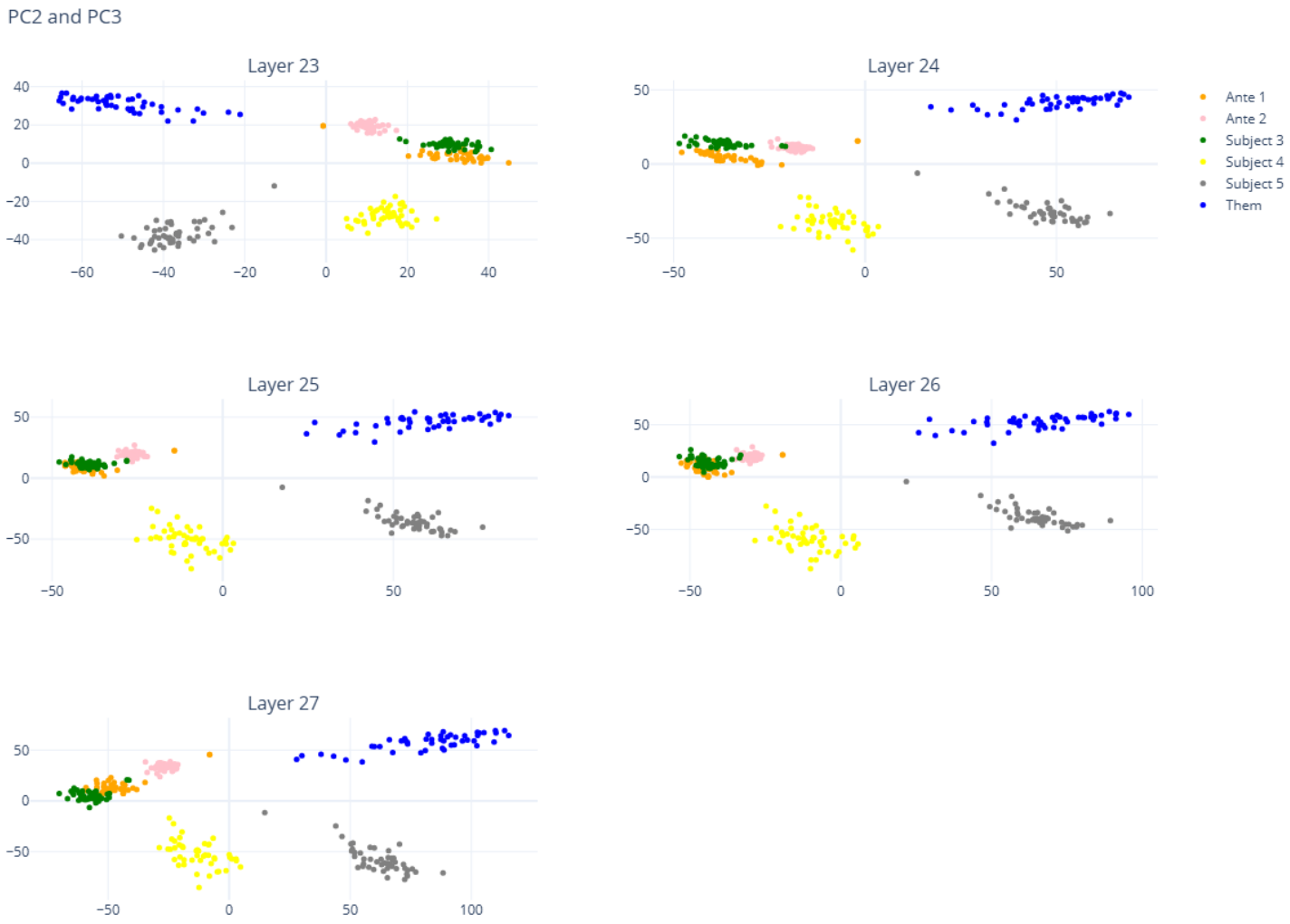
When John and Mary arrived at the restaurant that Nick and Paul owned, the waiter welcomed them

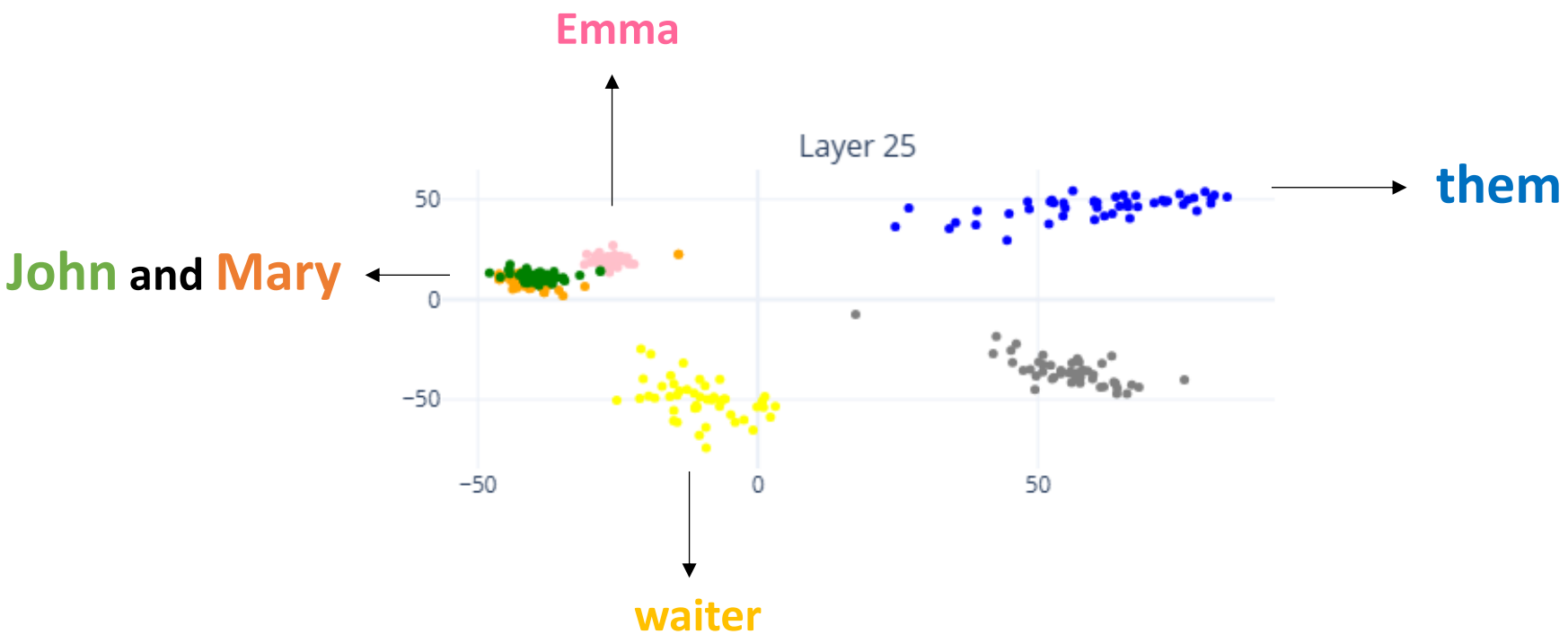


- More evidence on **how LLMs represent "groups"**

When **Emma**, **John** and **Mary** arrived at the restaurant that **Nick** owned, the **waiter** welcomed **them**

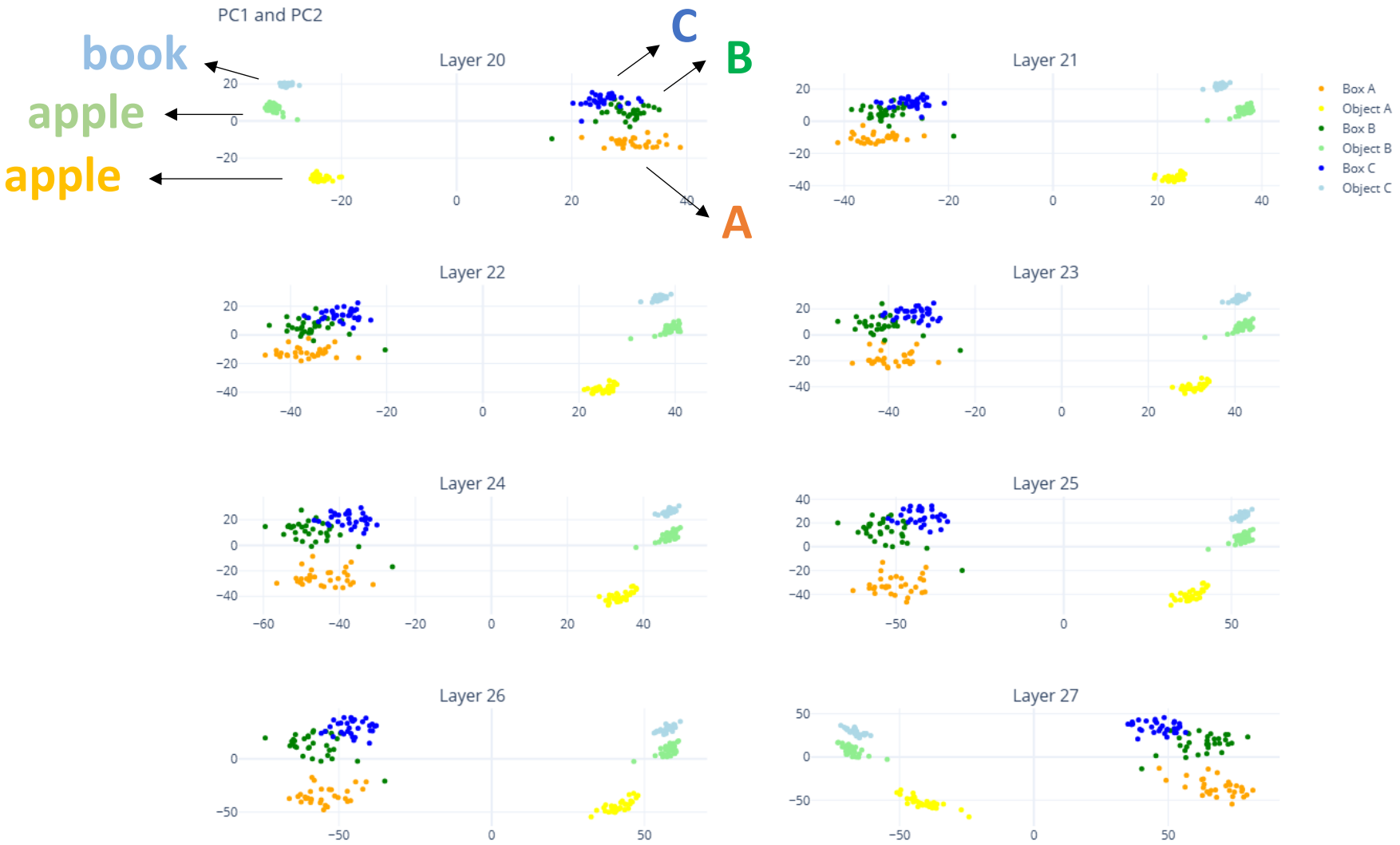


When Emma, John and Mary arrived at the restaurant that Nick owned, the waiter welcomed them

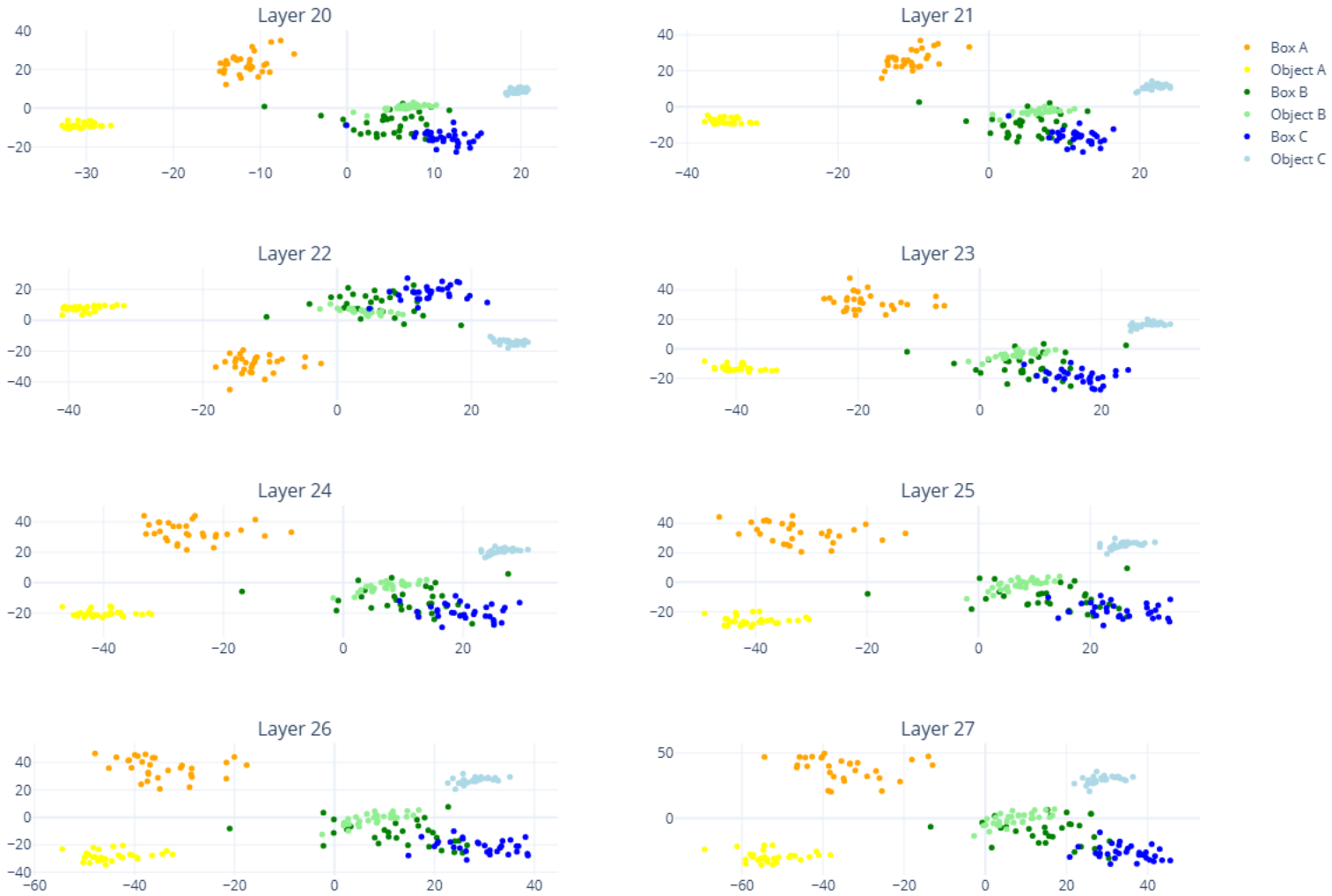


- **Connectives strongly affect group representation**

The coffee is in Box A. The apple is in Box B. The book is in Box C



PC1 and PC2


Layer 20
Layer 21
Layer 22
Layer 23
Layer 24
Layer 25
Layer 26
Layer 27
Box A
Object A
Box B
Object B
Box C
Object C